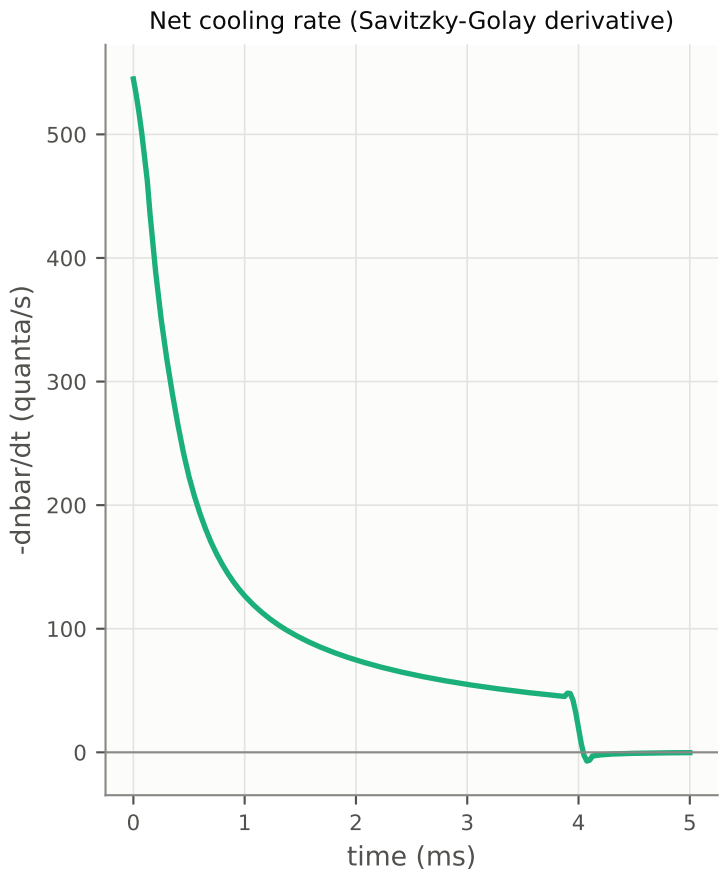
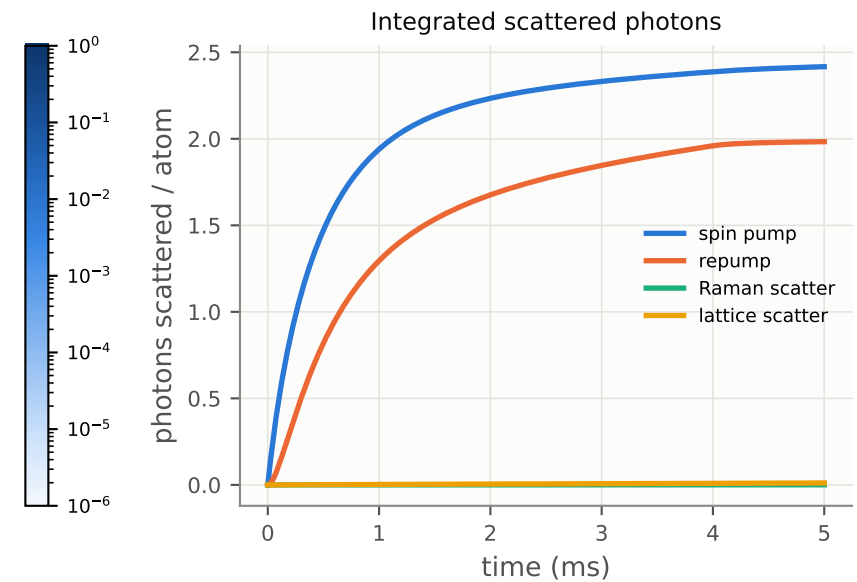
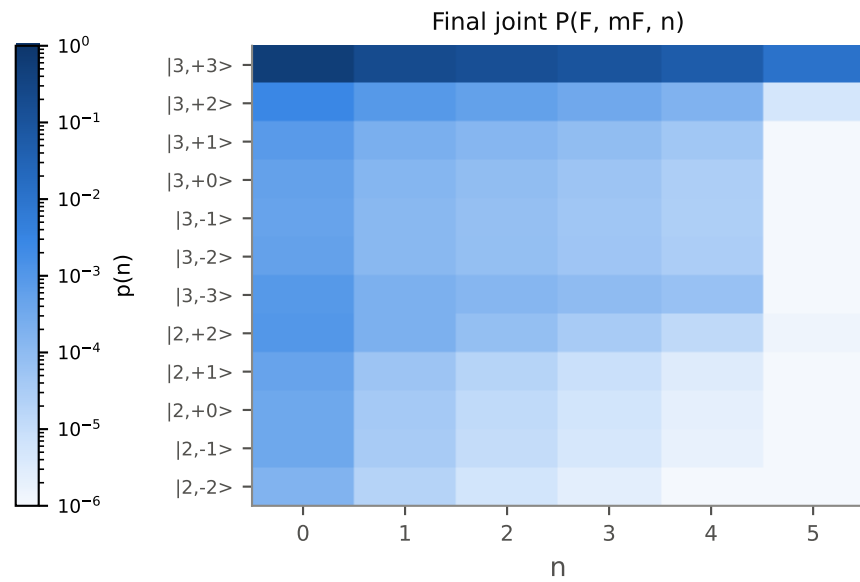
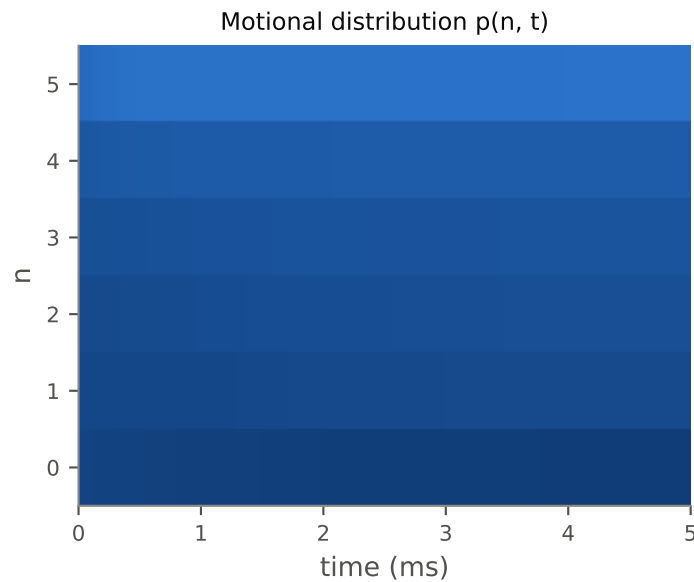
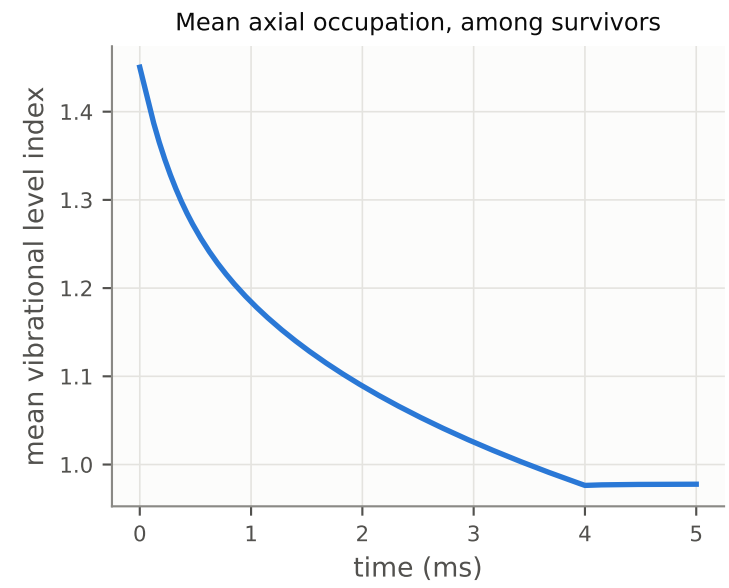
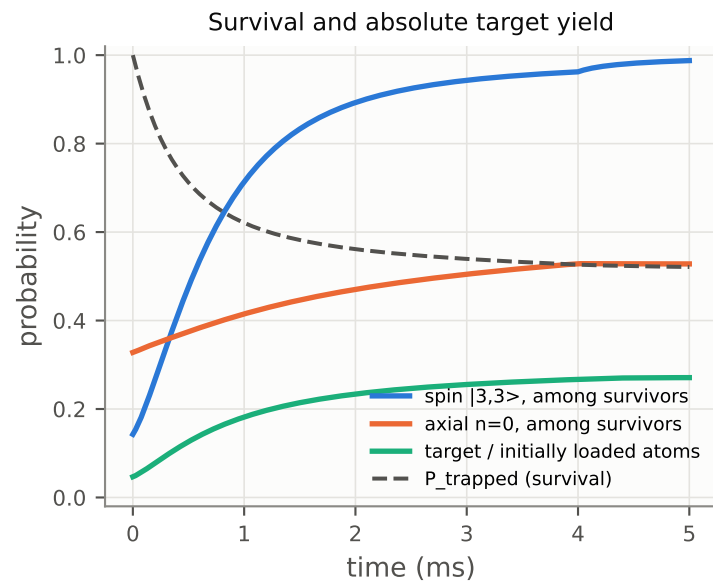
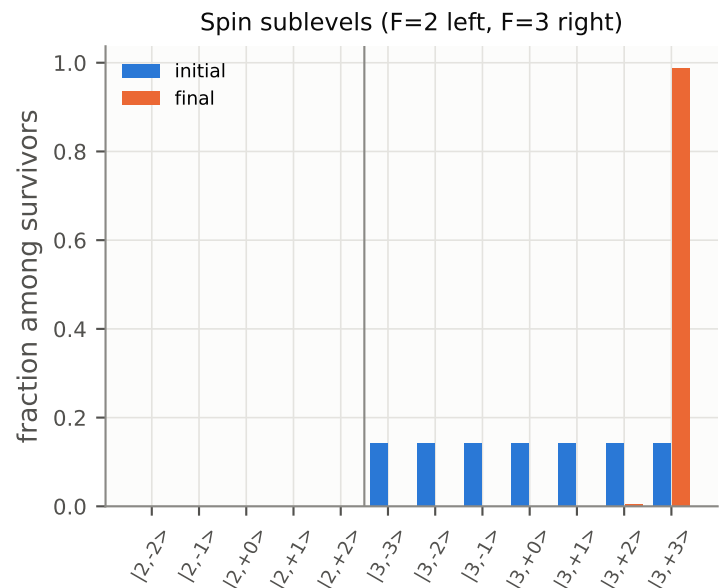
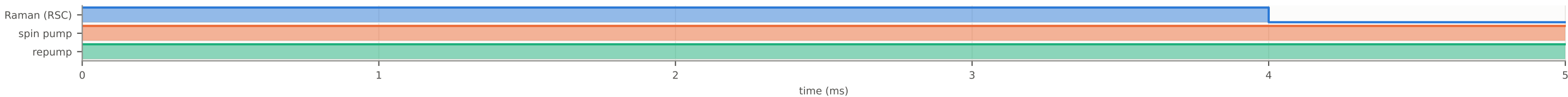


clarified_3d_experiment - protocol custom

Beams on (trace height = amplitude multiplier)



MODEL VALIDITY: WARNING
 $P_{\text{up}}=0.9877$ $P_{\text{n0}}=0.5280$ $P_{\text{target}}=0.5200$ ($P_{\text{up}}P_{\text{n0}}=0.5215$)
 populations are per trapped atom: trapped 0.5211 (lost 0.4789); of all atoms: $P_{\text{up}}=0.5146$ $P_{\text{n0}}=0.2751$ $P_{\text{target}}=0.2709$
 nbar (trapped atoms) 1.45 $\rightarrow$ 0.9777 $T_{\text{equiv}}=4.1$ uK [Energy-matched mixture of local bound canonical distributions, weighted by surviving spatial populations; proxy only. None means no finite positive-temperature match.]
 Well-loss fraction, assuming no recapture: 0.4789
 Axis temperature proxies (uK): $x=6.53$, $y=7.01$, $z=4.1$
 photons/atom: pump 2.42, repump 1.98, Raman 0.000539
 target $|3,3\rangle$ scattering rate 28.4 /s; quanta removed per photon None
 $t(P_{\text{up}}\geq 0.9) = 2.09$ ms; $t(P_{\text{target}}\geq 0.5) = 3.75$ ms
 conservation err $3.7\text{e-}14$; min eig 0; boundary pop unspecified; overflow 0.48
 solver coherent_3d_split_ensemble, wall 237.6 s

[OK] conservation: max $|\text{survival} + \text{well loss} - 1| = 3.69\text{e-}14$
 [OK] positivity: minimum density eigenvalue / population = 0
 [OK] weak excitation: maximum eliminated excited fraction across nodes and spin states = 0.000617
 [WARNING] motional secular: max optical rate / smallest local motional angular spacing across all axes = 0.48
 [OK] carrier calibration: Modeled spatial/thermal decay; RMS carrier frequency = 5000 Hz fixes initial curvature. No fitted homogeneous damping; reported decay is not a calibration constraint or an experimental validation.
 [WARNING] three-dimensional recoil: Joint spin/ $n_x/n_y/n_z$ populations retained, but emission uses resolved spherical dipole channels with factorized Cartesian marginals. Interference between emitted spherical channels and correlations of one photon's Cartesian recoil components are neglected.
 [WARNING] recoil coherence: Population-resolving scattering erases motional coherence and treats elastic spin amplitudes as distinguishable; this is an additional approximation beyond secular averaging.
 [WARNING] lattice: Separable isolated wells with Gaussian local depths; no tunnelling, coherent Raman continuum excitation, envelope transport or recapture. Scalar trap basis with diagonal differential shifts; trap shape changes from spin-dependent shifts neglected.
 [WARNING] Stark shift approximation: Raman/trap Stark operators use diagonal matrix elements; static off-diagonal motional dressing and trap-basis changes are omitted. Excited-state lattice shifts of the pumping transitions remain the configured common optical shift.
 [WARNING] spatial ensemble: 64 frozen Gaussian position / uniform registration samples. Refine samples and time_step_us for convergence; samples are numerical quadrature, not experimental error bars.
 [WARNING] temperature: Axis temperatures are energy-matched bound-state mixture proxies; transverse survivor cooling may result from preferential loss.